
ORBITAL INTELLIGENCE
GLOBAL CARBON SEQ: 1.2 MT
WATER CONSERVATION:
30%
45M
SMALLHOLDERS IMPACTED
JAPAC
REGION
10K+
MICRO-CLIMATES
CARBON SEQ: 1.2 MT
WATER CONSERVATION: 30%
RESILIENCE SCORE: 8.5/10
GLOBAN SEQ: 1.2 MT
WATER CONSERVATION: 30%
RESILENCE: 8.5/10
AgriResilience AI
Hyperlocal Multimodal Agronomy & Micro-Climate Defense for JAPAC Smallholders.
98%
OPTIMIZATION
2.5°C
REDUCTION
SOIL N: 120 PPM
WATER: 65% SATURATION
BIOMASS: +15% GROWTH
AI Builder Cup 2026 | Theme: Sustainability & Social Impact
Gemini Notebook


150° W
180° E
84°
76°
90°
26°
450M
Farmers
producing 80% of regional food supply
$30B
Annual Crop Loss
driven by 25–40% seasonal harvest write-offs
1:3,000
Extension Ratio
resulting in a 7 to 14 day advisory latency
The Agronomic Extension Desert.
Accelerating climate shocks breed aggressive blights. Farmers are trapped in an information void, relying on guesswork and predatory chemical dealers.
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Legacy Image Apps
(e.g., Plantix)



Blind to weather and soil conditions.



Text-heavy UIs (fails the 68% illiteracy barrier).



Pushes expensive branded chemicals.

Enterprise SaaS
(e.g., CropIn)



Built for massive estates and corporate FMCGs.



Coarse 10km satellite grids.



$5,000+ licenses, inaccessible to smallholders.

AgriResilience AI
(The Breakthrough)



Micro-weather aware via Google Earth Engine.

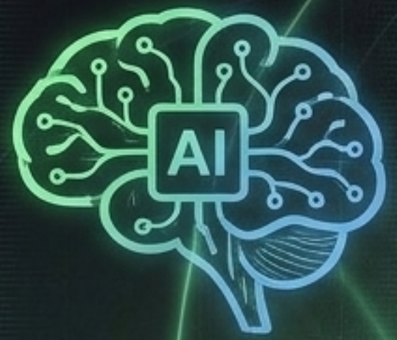


Vernacular Voice-First via Gemini 1.5 Pro.

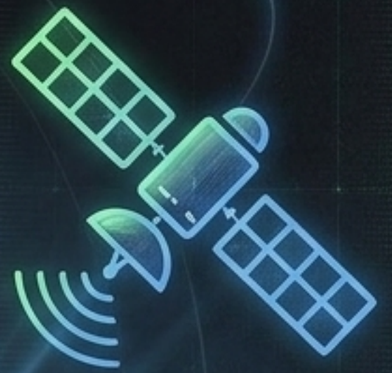


Prioritizes safe, low-cost organic recipes.

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Bridging the extension gap by placing an elite, multilingual plant pathologist and satellite meteorologist directly into the pocket of every smallholder.



Cognition:
Elite Plant Pathologist

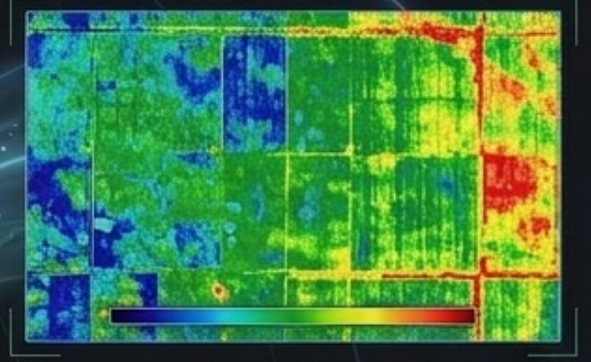
Meteorology:
Planetary-Scale Radar



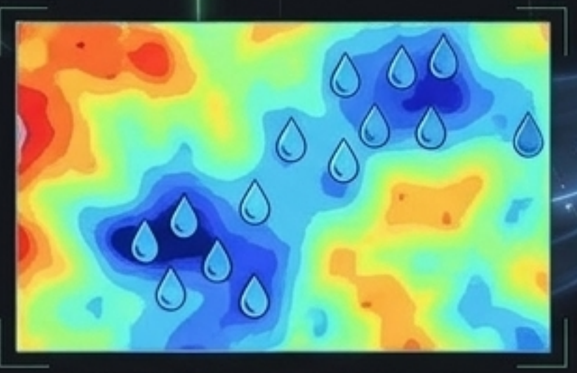
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Visual Lesion Markers



Satellite Vegetation Health (NDVI)



Soil Moisture (NDWI)

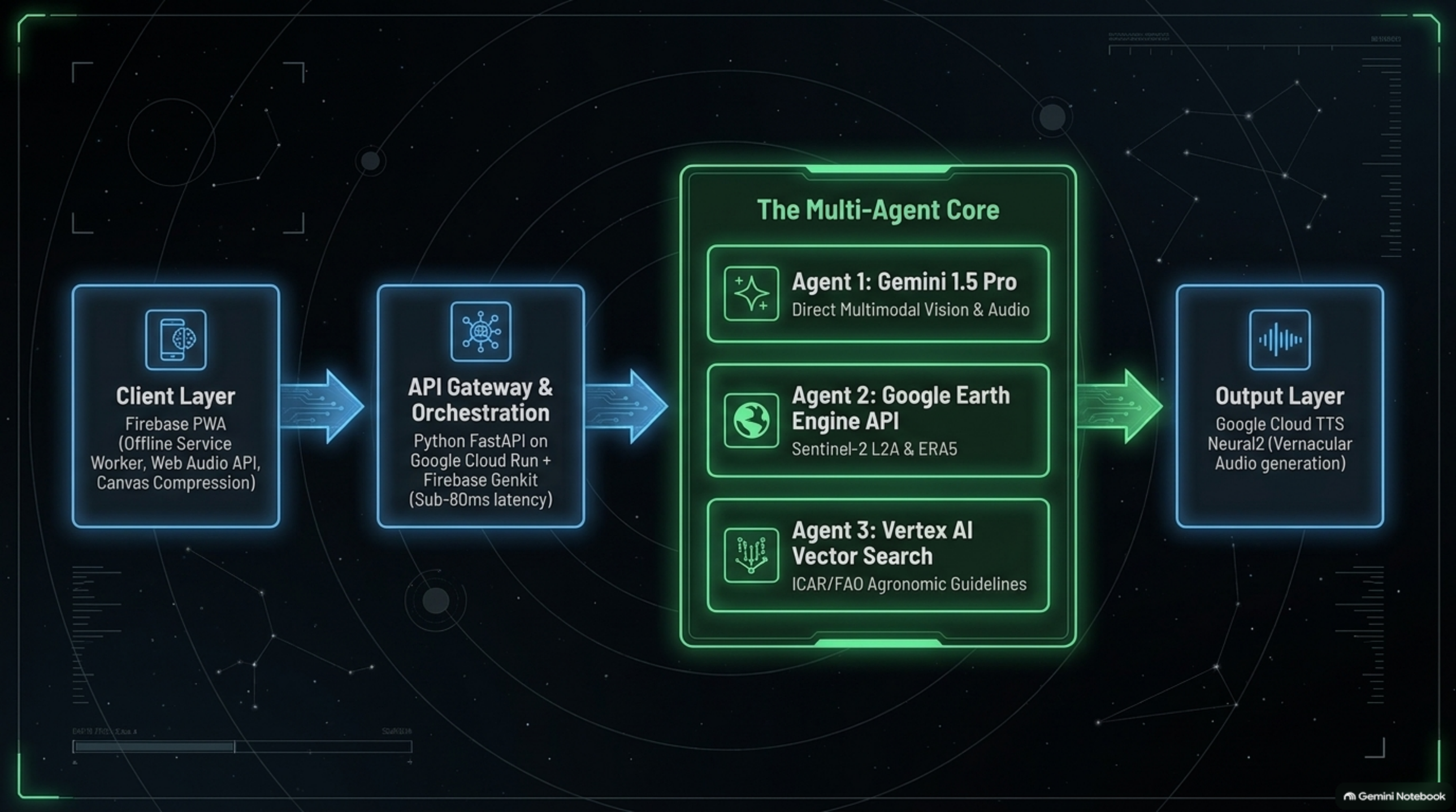


72h Precipitation (ERA5)

Prescriptive
Confidence = f(Pathogen,
Severity, NDVI, Rain Probability)

Legacy AI diagnoses a leaf in a vacuum. AgriResilience constructs an unprecedented, micro-climate aware Environmental Pathology Graph.

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Direct audio ingestion of native dialects (Marathi, Bahasa, Bengali)—no typing required.



Gemini natively processes the farmer's spoken distress level, crop identity, and exact fungal/bacterial sporulation stage in a single pass.

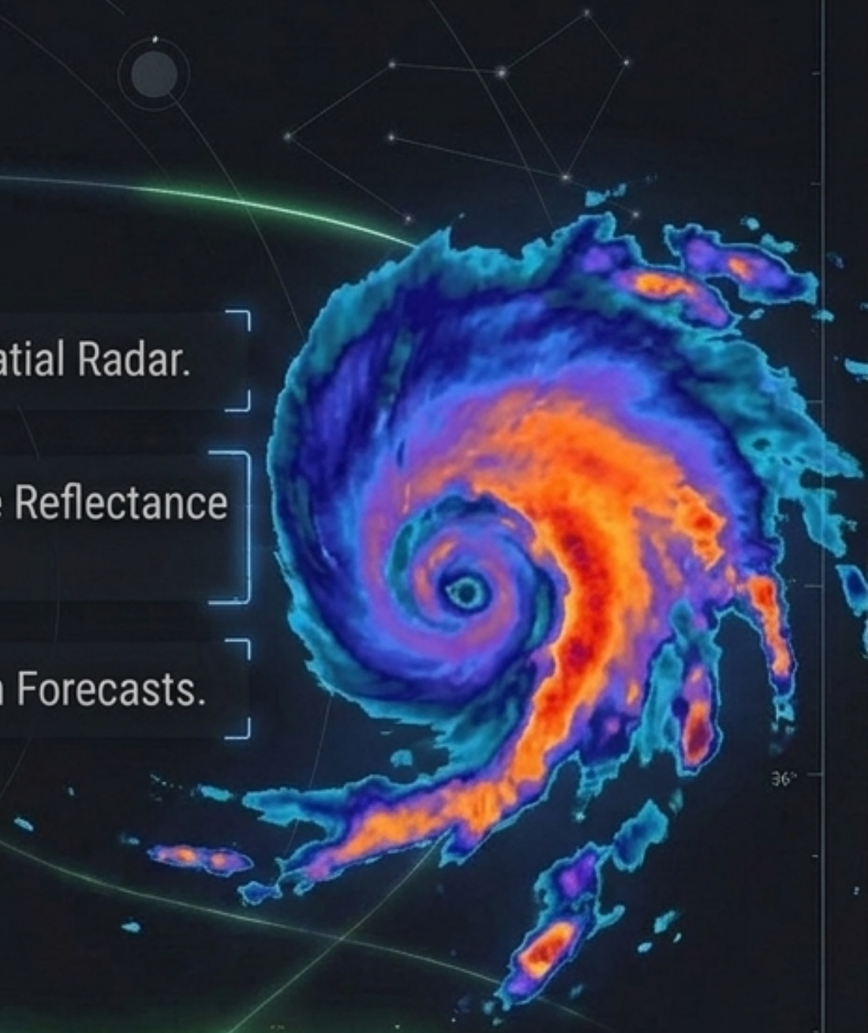
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Hyperlocal Geospatial Radar.

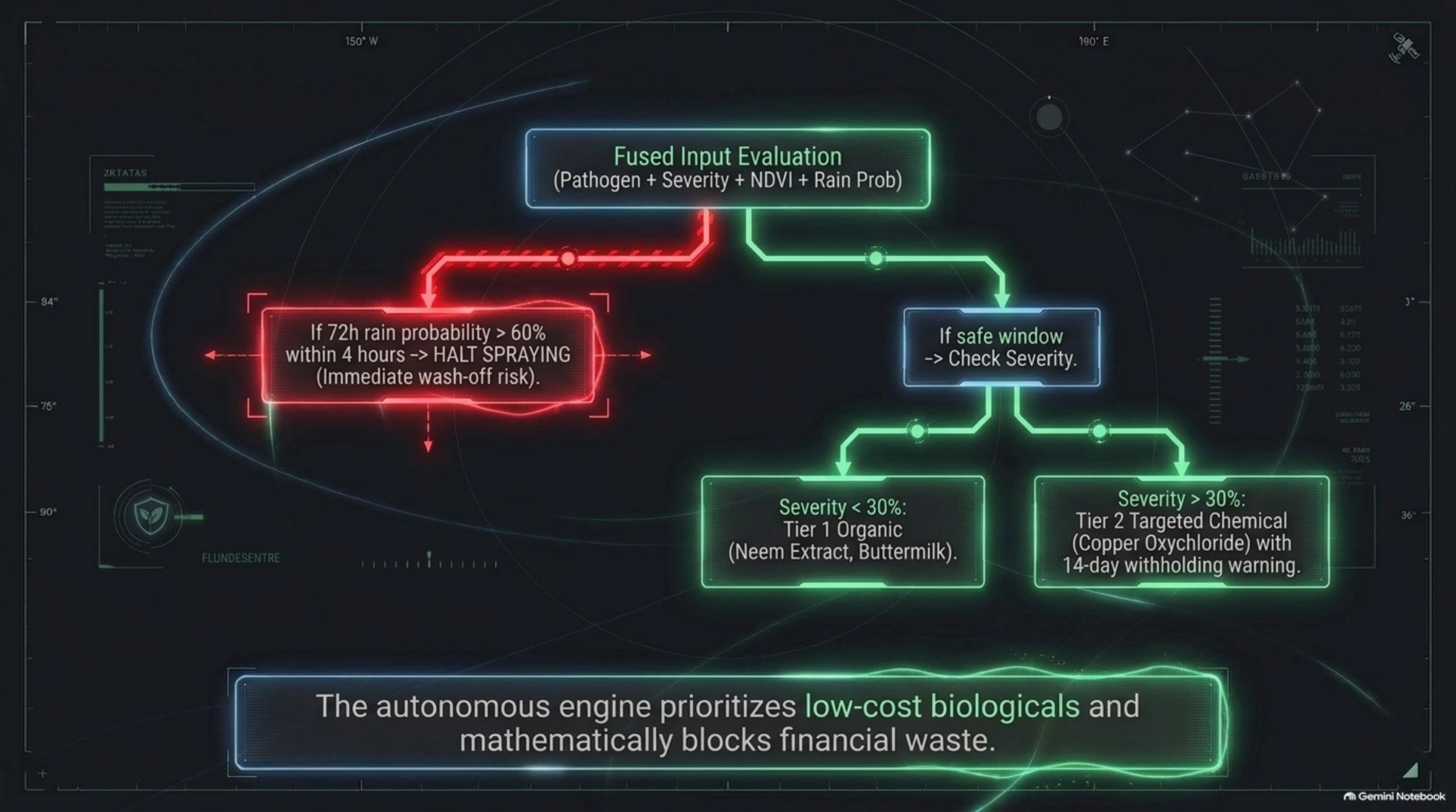
Sentinel-2 Surface Reflectance (10m resolution).

ERA5 Precipitation Forecasts.



Validates visual symptoms against actual canopy moisture stress and imminent weather events.

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Fused Input Evaluation
(Pathogen + Severity + NDVI + Rain Prob)
If 72h rain probability > 60%
within 4 hours -> HALT SPRAYING
(Immediate wash-off risk).
If safe window
-> Check Severity.
Severity < 30%:
Tier 1 Organic
(Neem Extract, Buttermilk).
Severity > 30%:
Tier 2 Targeted Chemical
(Copper Oxychloride) with
14-day withholding warning.
The autonomous engine prioritizes low-cost biologicals and
mathematically blocks financial waste.



0 1 TUKARAM

STEP 1

INPUT: Tukaram speaks in Marathi and uploads a photo of a wilting cotton leaf.

STEP 2

SYNTHESIS: Gemini spots Bacterial Blight (38%). Earth Engine detects an 18mm rainstorm arriving in 3.5 hours.

STEP 3

HALT

OUTPUT: The system flashes a red HALT warning, blocking chemical spray. It issues a spoken Marathi audio prescription to apply ₹45 organic neem extract tomorrow morning.

Saved ₹500 in wasted chemical inputs in exactly 8 seconds.

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COMMUNITY OUTBREAK RADAR (Epidemiological Scale)

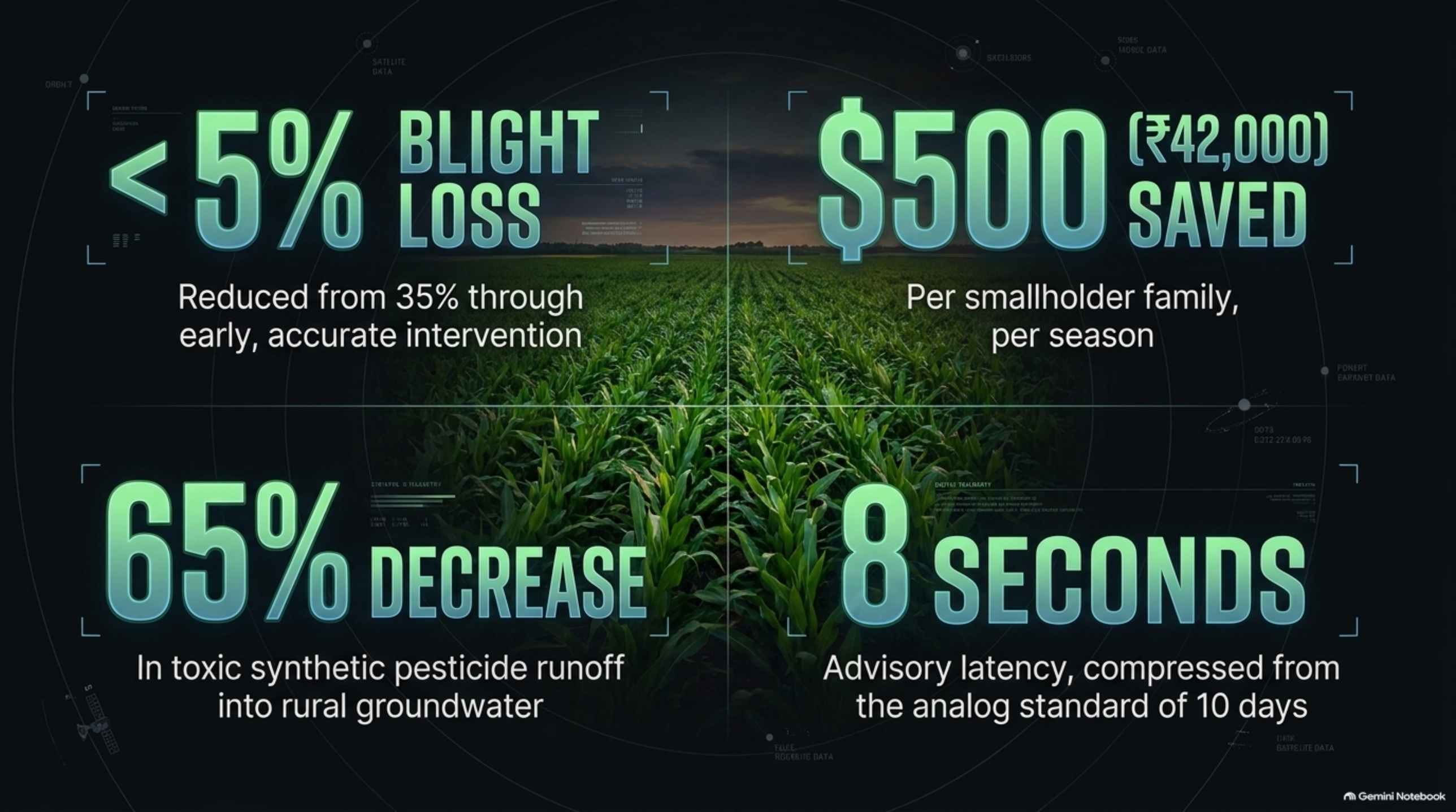


FROM INDIVIDUAL DIAGNOSIS TO REGIONAL DEFENSE.

WHEN 3+ NEIGHBORING FARMS CONFIRM AIRBORNE/WATERBORNE SPORES, THE EPIDEMIOLOGICAL CLUSTER ALERT TRIGGERS.

PROACTIVE PUSH NOTIFICATIONS WARN SURROUNDING FARMERS TO INSPECT BORDER ROWS BEFORE SPORES CAN SPREAD.

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Live Prototype Deployed.
Pre-packaged dossiers
ready for testing.



GitHub Repo URL
& Architecture
Documentation Link.

Bridging the Agronomy Gap. See you in Singapore.

AI Builder Cup 2026

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